

SUMMARY

I am a Ph.D. candidate in the Department of Earth and Environmental Sciences at Columbia University, advised by Dr. Jason Smerdon and Dr. Richard Seager. I graduated from Smith College in 2022 with a B.A. in Geosciences and Mathematics. During my undergraduate studies, I conducted honors research with Dr. Gregory de Wet, reconstructing regional temperatures from lake sediments at Darwin Crater, Tasmania, using the brGDGT biomarker paleotemperature proxy. At Columbia, my research explores how the ocean has contributed to Earth's climate in the past, present, and future. I am particularly interested in ocean–atmosphere interactions, modes of climate variability, and how the ocean responds to external forcings and, in turn, influences hydroclimate. My first chapter assessed whether observed long-term trends in tropical Pacific sea surface temperatures could be attributed solely to internal variability. Current projects focus on volcanic forcing, examining the ocean's response to large tropical eruptions and their climatic and hydroclimatic impacts. My career goal is to improve societal resilience to anthropogenic climate change by deepening understanding of the ocean's role in long-term climate variability and change, which I believe is key to better near-term climate projections.

Research topics: ocean–atmosphere interaction, long-term climate variability and change, paleoclimatology

EDUCATION

Columbia University/Lamont–Doherty Earth Observatory

New York, NY

Doctor of Philosophy in Earth & Environmental Sciences

Expected: Summer 2027

Master of Philosophy in Earth & Environmental Sciences

February 2026

Master of Art in Earth & Environmental Sciences

May 2024

- Advisors: Dr. Richard Seager, Dr. Jason Smerdon, and Dr. Lorenzo Polvani

Smith College

Northampton, MA

Bachelor of Art in Geosciences with Highest Honors & in Mathematics

January 2022

- Honors Thesis: 'Reconstructing Past Temperature from Lake Sediments at Darwin Crater, Tasmania, Australia Using the brGDGT Biomarker Proxy' – advised by Dr. Gregory de Wet
- *Cum laude* (GPA: 3.91 out of 4.0)

RESEARCH EXPERIENCE

Dissertation Research

New York, NY | Fall 2024–Present

Columbia University/Lamont–Doherty Earth Observatory

Project title: 'A Comprehensive Assessment of Superposed Epoch Analysis for Estimating Global Patterns of Volcanic Impacts in Climate Field Reconstructions'

- Examined the sensitivity of spatial patterns of volcanic forcing signals derived by Superposed Epoch Analysis to methodological choices and data uncertainties using paleo data assimilation products.
- Manuscript is currently in preparation.

Master's Thesis Research

New York, NY | Spring 2023–Present

Columbia University/Lamont–Doherty Earth Observatory

Thesis title: 'A Comprehensive Assessment of Observed Trends in the Zonal Sea Surface Temperature Gradient in the Tropical Pacific Ocean'

- Investigated the observed trends in the equatorial Pacific's zonal sea surface temperature (SST) gradient over 20-year or longer intervals within the 1870–2024 period using regrided SST datasets.
- Evaluated these observed trends against a series of null hypotheses using bootstrapped ensembles of various statistical (white and red noise), conceptual, and geophysical models.

RESEARCH EXPERIENCE (continued)

NASA DEVELOP National Program – Intern

Remote | January–April 2022

Space Systems And Innovations Laboratory Limited Co.

Project title: ‘Mapping Forest Carbon Stocks to Understand Carbon Implications of Treatment and Wildfire’

- Main advisors: Dr. Tony Vorster, Dr. Nicholas Young, Dr. Brian Buma, and Scott Cunningham
- Used machine learning (random forest) on RStudio to approximate forest carbon pool sizes of the burned and surrounding unburned areas after the 2020 CalWood fire in Boulder and Jefferson Counties, CO, from field-based plot data, pre- and post-fire satellite imagery, and other remote sensing data.

Undergraduate Honors Thesis Research

Northampton, MA | February 2021–January 2022

Smith College

Thesis title: ‘Reconstructing Past Temperature from Lake Sediments at Darwin Crater, Tasmania, Australia Using the brGDGT Biomarker Proxy’

- Advisor: Dr. Gregory de Wet
- Estimated past Tasmanian annual mean air temperatures from the lake sediment samples using branched glycerol dialkyl glycerol tetraether (brGDGT) as the paleotemperature proxy to understand past regional environmental conditions and climate (presumably in MIS 5c-6).
- Conducted highly complex biogeochemical laboratory work to analyze the sediment samples.

Marine Biology Research – Summer Intern

Gloucester, MA | June–August 2019

University of Massachusetts Amherst’s Gloucester Marine Station

Thesis title: ‘Evolutionary response to climate change on US coastal shores’

- Assisted in master’s level research on the potential impacts of climate change on evolutionary processes in marine snails (*Urosalpinx cinerea*) for 10 weeks.
- Work involved care and husbandry of roughly 200 adult and over 1000 juvenile marine snails, maintenance of 9 seawater tanks, photographically recording the growth of juvenile snails with a microscope using Image J, and data management using Microsoft Excel.

Independent Geosciences Research

Northampton, MA | January–May 2019

Smith College

Project title: ‘Exploring the influence of seasonality on earthquake occurrence in Japan’

- Main advisors: Dr. Jack Loveless
- Investigated the seasonal patterns in earthquakes by exploring the potential relationship between the snowfall amounts and frequencies of the earthquake occurrence in Japan.
- Independently constructed the project, self-taught MATLAB, and performed statistical analyses.

Coastal Science Research – Project Member

Northampton, MA | January–May 2018

Smith College

Project title: ‘Using Google Earth historical images to assess the impact of hurricanes in the Bahamas’

- Main advisors: Dr. Bosiljka Glumac
- Participated in *Early Research Program* administered by the AEMES (Achieving Excellence in Math, Engineering and Sciences) initiative at Smith College.
- Collaborated with seven peer students on a project exploring the ability of Google Earth images to describe the hurricane damage on coastlines of the Bahamas.
- Coauthored a conference abstract from the Annual National Geological Society of America Meeting in Indianapolis, IN in November 2018.
- Gave a poster presentation at the Smith College Collaboration Conference on April 21, 2018.

TEACHING EXPERIENCE

Columbia University

New York, NY

Teaching Assistant – *Earth's Environmental Systems: Climate Systems*

January–May 2025

- Hosted 1-hour weekly office hours.
- Graded lab reports and homework assignments.
- Led weekly ~3-hour lab sessions with six other teaching assistants; the role included designing and preparing hands-on and computer-based exercises, as well as providing in-class instruction.

Teaching Assistant – *Oceanography*

September–December 2023

- Hosted 1-hour weekly office hours.
- Worked together with teaching assistants and course instructors to develop and update bi-weekly homework assignments, midterm, and final exams.

Smith College

Northampton, MA

Tutor – Japanese

September–December 2021

September 2018–December 2019

- Tutored introductory (100-level) and advanced (300-level) Japanese courses, with an emphasis on oral communication skills.
- Conducted weekly 30-minute tutoring sessions with one or two students, totaling 4–5 hours per week.
- Independently developed session materials.

PUBLICATIONS & CONFERENCE

- Sugiura, I.**, Smerdon, J.E., and Serger, R. (2026) ‘Observed Trends in the Tropical Pacific Zonal Sea Surface Temperature Gradient Are Statistically Significant Based on Multiple Null Hypotheses’, *J. Climate*, 39, 4769–4787, <https://doi.org/10.1175/JCLI-D-25-0460.1>.
- He, Z., Tejedor, E., Smerdon, J.E., Vuille, M., Polvani, L.M., Seager, R., and **Sugiura, I.** (2026) ‘Comparison of Global Climatic Responses to Large Tropical Volcanic Eruptions over the Last Millennium in Paleoclimatic Reconstructions and Model Simulations’, *J. Climate*, 39, 1295–1313, <https://doi.org/10.1175/JCLI-D-25-0179.1>.
- Sugiura, I.**, Smerdon, J.E., Tejedor, E., Vuille, M., and Polvani, L.M. ‘A Comprehensive Assessment of Superposed Epoch Analysis for Estimating Global Patterns of Volcanic Impacts in Climate Field Reconstructions’, The American Geophysical Union 2025 Fall Meeting Oral Session, New Orleans, LA.
- Sugiura, I.**, Seager, R., and Smerdon, J.E. (2024) ‘Centennial Trends in the Tropical Pacific Zonal Sea Surface Temperature Gradient Are Outside the Range of Multiple Significance Tests’, The American Geophysical Union 2024 Fall Meeting Poster Session, Washington D.C..
- Bryne, H., Smerdon, J.E., Seager, R., and **Sugiura, I.** (2024) ‘Strengthening Trends in the Zonal Gradient of Tropical Pacific Sea Surface Temperatures Are Not Captured by CMIP6 Large Ensembles When the 21st Century Is Included’, The American Geophysical Union 2024 Fall Meeting Poster Session.
- He, Z., Tejedor, E., Smerdon, J.E., Vuille, M., Polvani, L., **Sugiura, I.**, and Seager, R. (2024) ‘Comparison of Global Climatic Responses to Large Tropical Pacific Volcanic Eruptions Over the Last Millennium Paleoclimatic Reconstructions and Model Simulations’, The American Geophysical Union 2024 Fall Meeting Online Poster Session, Washington D.C..
- Sugiura, I.**, de Wet, G., Lisa-Pronovost, A., Fletcher, M.S., and Cooley, S. (2022) ‘Reconstructing Past Temperature From Lake Sediments in Darwin Crater, Tasmania, Australia Using the brGDGT Biomarker Proxy’, The Geological Society of America 2022 Northeastern Section Meeting Poster Technical Session, Lancaster, PA.

PUBLICATIONS & CONFERENCE (continued)

- Hettema, S., Rogers, J., **Sugiura, I.**, Twaddell, E., Cunningham, S., Vorster, A., Buma, B. (2022) ‘Mapping Forest Carbon Stocks to Understand Carbon Implications of Treatment on Wildfire for the Calwood Fire, Boulder County Colorado’, The American Geophysical Union 2022 Fall Meeting Poster Session, Chicago IL.
- Miguel, U., Grow, A., Hinnert, J., Leiser, L., Lu, K., Mutka, M., Papineau, E., **Sugiura, I.**, Glumac, B., and Curran, H.A. (2018) ‘Using Google Earth historical images to assess the impact of hurricanes in the Bahamas’, The Geological Society of America Abstracts with Programs, Annual Meeting, Indianapolis, IN, v. 50(6).

AWARDS & HONORS

Columbia University	New York, NY
Fall 2025–Spring 2026	Teaching Assessment fellow at the Center for Teaching and Learning
Fall 2024–Spring 2025	Teaching with AI fellow at the Center for Teaching and Learning
Fall 2022	Dean’s Fellowship
Smith College	Northampton, MA
May 2022	• Elected to the Smith College Chapter of Phi Beta Kappa. • DAAD (German Academic Exchange Service) graduate scholarship (declined)
April 2021	Nominated for associate membership in Sigma Xi.
Summer 2020 & 2021	DAAD’s Research Internships in Science and Engineering (both cancelled due to the COVID-19 pandemic)
Summer 2019	• Smith College’s Praxis scholarship • Five College Coastal & Marine Sciences summer internship scholarship
Fall 2018–Spring 2019	Dean’s List

OTHER RELEVANT EXPERIENCE

PaleoCAMP <i>Sierra Nevada Aquatic Research Laboratory</i>	Mammoth Lakes, CA July 2025
CESM Tutorial <i>National Center for Atmospheric Research MESA Lab</i>	Boulder, CO August 2024
Urbino Summer School in Paleoclimatology <i>University of Urbino</i>	Urbino, Italy July 2024
URSSI Summer School 2024 <i>George Washington University</i>	Washington D.C. June 2024
OceanHackWeek 2021 <i>Bigelow Laboratory for Ocean Sciences</i>	East Boothbay, ME August 2021

EXTRACURRICULAR ACTIVITIES

Columbia University/Lamont–Doherty Earth Observatory

New York, NY

Departmental Graduate Student Committee

Co-Chair

August 2025–August 2026

Social Chair & Postdoc Liaison Chair

August 2024–May 2025

First-Year Chair

August 2023–May 2024

- Surveyed students' interests and needs within the Department of Earth and Environmental Sciences, and hosted community-building and served events based on such survey results.
- Planned and facilitated events for the PhD student community, such as a lunch event with postdocs to learn about postdoctoral opportunities and their application processes.
- Initiated a campus-wide student survey to hear students' opinions on their office conditions.
- Participated in several faculty meetings and advocated for international students' need for summer school and scholarship opportunity information that are not limited to domestic students.

Ocean & Climate Physics (OCP) Divisional Seminar Organizing Committee

Member

August 2025–May 2026

- Organized and facilitated the OCP divisional seminar series at the Lamont–Doherty Earth Observatory.
- Responsibilities included creating nomination forms for speakers, researching and contacting potential speakers via email, and managing logistical details such as scheduling talks, coordinating the speakers' agendas for visit days, and arranging one-on-one meetings.

Colloquium Organizing Committee

August 2023–May 2025

Division Representative

- Organized and coordinated the Earth science colloquium series for the Lamont–Doherty Earth Observatory community.
- Served on the Ocean & Physics Division representative in the organizing committee with five student representatives and a faculty advisor.
- Responsibilities included identifying and contacting potential speakers, managing scheduling, and arranging 1-hour lunch events for speakers and graduate students.

DEES Application Mentoring Program

Founding Member & Mentor

August 2023–January 2024

- A mentoring program for prospective applicants to the PhD programs in the Department of Earth & Environmental Sciences (DEES). The goal is to promote diversity by supporting applicants from underrepresented groups. The prospective students are paired with current PhD students.
- Website: <https://eesc.columbia.edu/content/dees-application-mentorship-program-dees-amp>
- Established this mentoring program with another PhD student in Fall 2024.
- During the 2025 admission cycle, we helped 53 mentees.

XPLANE

Online

Graduate School Application Mentoring Program

Mentor

September 2022–April 2025

- XPLANE is a non-profit organization that offers a community over the Slack platform for Japanese graduate students who are studying abroad.
- Volunteering as a mentor in the graduate school application mentoring program – advising Japanese students who are interested in pursuing graduate degree abroad, mainly by giving tips how to choose and contact potential advisors, providing language support if necessary, and reviewing their application essays. I mentored three students, who all have been accepted to graduate programs in US and Europe.

EXTRACURRICULAR ACTIVITIES (continued)

Smith College

Northampton, MA

International Student Organization (ISO)

Publicity Chair

June 2021–December 2021

Cultural Chair

February–June 2021

Publicity Chair

April 2019–December 2019

Social Chair

April 2018–April 2019

- ISO is a student organization dedicated to fostering a community of international students.
- Managed logistical arrangements for social events and designed digital and physical posters to advertise those events.
- Co-led the *My Global Story* project, providing space for international students to share their experiences in the USA through videos, poetry, and other artworks. The project received the Smith President's Innovation Grant and was supported by the International Students' and Scholars' Office and the Narrative Project.

Japanese Student Association (JSA)

President

September 2019–December 2019

Vice President

March 2018–May 2019

- Founded and led JSA, a student organization providing a supportive community for students of Japanese backgrounds and offering peer mentorship programs.
- Established the organization with original members in Spring 2018.

SKILLS

Programming

- Proficient: R/RStudio, Python
- Familiar: Java, MATLAB

Languages

- Japanese (native)
- English (fluent)

ArcMAP (ArcGIS Software)

- Experienced with data input and output, processing, and analysis.
- Analyzed watershed and its surrounding geochemical and biological features.
- Generated watershed maps showing those biogeochemical variables.